

from digital signals and floating-point calculations. Due to the current supply limitation, designers do not have the power to boost processing speeds by raising the CPU clock. The resulting long processing periods increase the power loss for the entire system.

The new STM32L0 ultra-low-power series from STMicroelectronics features a 32MHz ARM Cortex M0+ processor core and a 12-bit analogue-to-digital converter (ADC). "This new standard-bearer enables customers to design more competitive products that will both bring immediate cost and convenience benefits to end users and reduce worldwide consumption of disposable batteries," said Vinay Thapliyal, MCUs marketing and technical manager, STMicroelectronics. With the ADC consuming just 40µA at a conversion speed of 100ksp/s, rising to just 200µA at 1.14Mps, in combination with the outstanding power efficiency of both the processor core and ultra-low-power peripherals, applications such as fitness trackers and domestic smoke detectors powered by non-rechargeable batteries will require less frequent replacements, saving both end user costs and the environmental burden of spent batteries.

The new RX231 group is for applications that require high processing performance in environments with low current supply capacity, with digital signal processing (DSP) and floating point unit. According to Renesas, it consumes only taking 0.8µA in standby mode while retaining the contents of both SRAM and CPU registers, and 120µA/Hz when the CPU is active with peripherals off.

Atmel's new SMART 32-bit SAM L21 featuring an ARM Cortex-M0+ core consumes just one-third the power of comparable products in the market today (according to Atmel). How? Their SMART SAM G54/G55 has multiple clock domains to enable different parts of the MCU—the core, internal buses and individual peripherals—to be clocked individually to save power. If the maximum throughput of a peripheral is not needed

in an application, the clock can be lowered. Individual clocks can be applied to different peripherals so these can be fully optimised with power consumption in mind. Throughput or feature requirement of the application is still fulfilled while power consumption is lowered.

Once Microchip's PIC16F18877 is initialised in a system, core-independent peripherals can provide steady-state closed-loop embedded control with zero intervention from the MCU's core. The CPU can then be idled or put into SLEEP mode to save system power. Microchip claims that PIC16F18877 is also the first PIC16 MCU to leverage Microchip's eXtreme Low Power (XLP) technology with IDLE and DOZE modes for reduced active power consumption. Members of this family are ideally suited for applications that involve LED lighting and motor control like for wind direction, fan and flow control.

Security, now a base ingredient

In the wake of recent hacks like the Chrysler Jeep situation, the connected critical systems require security to be a base ingredient. SPC58NE is one among the first chips targeting automotive space, to come out on STMicroelectronics' 40nm process, where security now extends to the hardware security module system, targeting E-safety Vehicle Intrusion Protected Applications (EVITA) Medium Class, which can be seen as an autonomous and isolated system, embedded into the MCU, handling all security operations.

For enhanced security, the RX231 group of MCUs provides AES encryption, a hardware based random number generator and unique device ID that can be configured to support advanced cryptography requirements.

The new automotive-grade S32K MCU lends SafeAssure functional safety to cars, which makes it easier to achieve system compliance with functional safety standards in automotive and industrial markets, while reducing the time taken to develop safety

systems that comply with ISO 26262 and IEC 61508 standards. The MCUs also incorporate a Secure Hardware Extension compliant module that helps to secure communications and protect the integrity of sub-systems.

Atmel's WINC1500 has an internal flash memory hosting a secure IoT network stack that can be controlled from any resource-constrained MCU, thanks to its very small footprint BSD socket API (4kB flash/1kB RAM). Implemented in 65nm CMOS technology, WILC1000 and WINC1500 offer very low power consumption, while simultaneously providing high performance with an optimised bill of material (BOM).

Broadcom's BCM58100 comes with BroadSAFE architecture, which offers tamper protection and encryption as well as secure storage and processing of both card information and biometric user data such as fingerprints, irises, facial templates and more to safeguard against attacks. BCM58100 is compliant with Federal Information Processing Standard (FIPS) and also ensures that end products achieve stringent Payment Card Industry and Europay, MasterCard and Visa certifications for an added layer of security.

More support for communication interfaces

"Features for integrating into the IoT are also another area. For example, ARM Cortex-M7 is optimised for the IoT. You have support for real-time clocks since time-stamping is very important in the IoT. It was earlier done externally through SPI protocol. Communication interfaces in the wireless domain like Wi-Fi and Zigbee are being shown, too," says Hemant Kamat, CTO at Shalaka Technologies.

"Many years back, we only had controller area network (CAN) integrated within the MCU. Nowadays, we have CAN, LIN and FlexRay integrated. Moreover, while we previously had to extend the number of CAN nodes by using external CAN, we can now do so from within the chip by making use of the multiple channels available.